M2-Act1: Tangrams (Individual) FJ26

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Clase

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Tangram Experiment: Hypothesis Tests Revisited

Abstract

Description

This experiment is an exemplary study for revisiting (one-sample) hypothesis testing.
We aim to find out whether the subjects can complete a tangram puzzle in a specific amount of time.

Tangram is historically a Chinese game, where one is given seven polygonal pieces (called *tans*[1]). The goal is to arrange these tans in such a way that a given shape has to be completed, while overlapping tans are not allowed. Below an example is given:

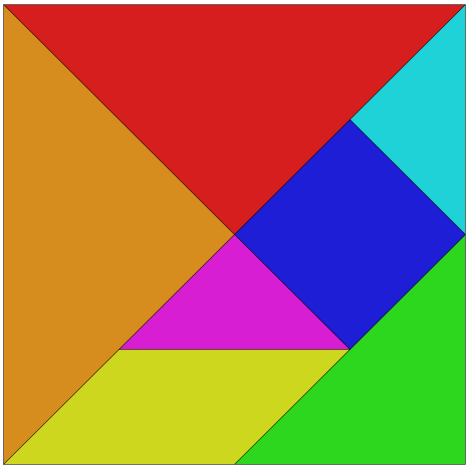

Figure 1. Example tangram.[2]

The different colours are used to distinguish the seven tans from one another.

Methodology

To conduct our experiment, we rely on the website *stat2games*, where the subjects each enter a different player ID, but the same group ID, in order to differentiate between them and other simultaneous users of the website. Our subjects are 11 students of the course.

A one-time session is held, where each subject plays the game on his own device, such that conditions are as homogeneous as possible for each participant. We disallow the use of hints. All further analysis is based upon the results of this session. Specifically, the completion time of all the subjects for the specific tangram shape "*House of Tangrams*" is measured by the website.

[1] https://www.brainzilla.com/puzzles/tangram/
[2] https://www.picuino.com/en/_images/taller-tangram-03.png

Objective

As briefly stated before, our only aim is to investigate whether the subjects can complete the "*Diamond*" tangram in a given amount of time. Specifically, it is our goal to determine whether the students can complete the tangram in under 2 minutes.
That is, whether their mean completion time remains under 120 s, as the completion time is measured in seconds.

Results

We could not verify that the completion time of the students for the tangram shape is under 2 minutes.

1 Establishing the Variable

First of all, we are measuring the completion time for the specific tangram shape *House of Tangrams*, in seconds. We are interested in the mean of this random variable.
Therefore, one could initially conclude that we shall use a normal distribution. However, as our sample size is way below 30, the student's t-test will be used.

Data Processing

When taking the sample data, filtering for runs of the correct shape (*House of Tangrams*), only keeping the first attempt in case of duplicates and removing attempts which were not won or which used hints, one is left with 11 samples. This is the data captured, sorted ascendingly:

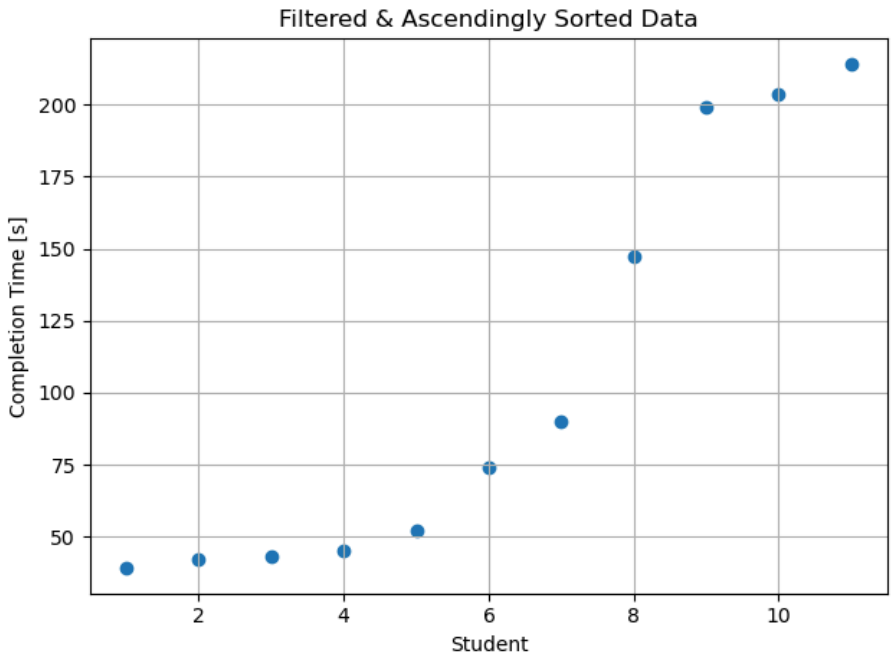

Figure 2. Captured data, filtered and sorted ascendingly.

Evidently, there are no outliers and we therefore do not need to perform any further cleaning of the gathered data.

Conditions

The study was conducted in the class room CIAP 417, at around 12 PM, on 26 March 2026. Beforehand, the students had about five minutes to accustom themselves to the tangram game, and practise with shapes other than *House of Tangrams*.
For the final run, each student was only allowed to try the game once and without hints.

During the first two minutes of the run, it was mostly quiet in the room. Thereafter, the noise level noticeably increased.

It is important to ensure the same conditions for each subject, so as to best possibly only capture the sole variable we want to measure for each student, the completion time.
If the conditions were not the same for each student, we could not unequivocally attribute the differences in the completion time only to their ability.

2 Establishing the Hypotheses

As we want to establish that the students can solve the specified tangram shape in under 2 minutes, this assertion determines our alternative hypothesis. Logically, the null hypothesis is its mere logical negation, namely the mean of the completion time being at least 2 minutes:

- H_0: $\mu \geq \mu_0$
- H_1: $\mu < \mu_0$

In this instance, since our measurements are taken in seconds, we have $\mu_0 = 120$.

3 Determining the Correct Distribution

As mentioned before, our sample size is very small and we do not know σ. Hence, we shall use the t-distribution, for which the formula is given in the cheat sheet on canvas:

$$t = \frac{\bar{x} - \mu_0}{s} \cdot \sqrt{n}$$

4 Calculating

We get the following results:

- $\bar{x} \approx 104.58$
- $s \approx 72.02$
- $t \approx -0.7102$ (using Excel)

Therefore, using the student's t-distribution, we obtain the following values for p and critical t:

- $t_c \approx -1.812$

- $p \approx 0.2469$

5 Conclusion

Evidently, as $|t| < |t_c|$, we can not reject the null hypothesis and therefore have no sufficient evidence to establish the alternative hypothesis.
Equivalently, as $\alpha = 0.05$ is given in the task statement, one can also see that $p \geq \alpha$, leading to the same conclusion.

In another way, looking at the 95% confidence interval for the mean $(56.44,\ 152.72)\ s$, we can clearly see that μ_0 lies within this interval.
Regarding point 10, this confidence interval means that we are 95% confident that the true mean to complete the tangram lies within it.
The utterance "we are 95% confident that each student takes between a and b seconds to complete the tangram" is trivially falsified as there are multiple students which lie outside this interval, which is already known a priori.
The last statement, "95% of students are expected to complete the tangram in a to b seconds", is also wrong, because the confidence interval only states a property about the mean, but not about individual values. We can not make this statement a confidence statement about individual values.

6 Interpretation

Due to to the high sample standard deviation (recall, $s \approx 72.01$), we can not establish sufficient evidence to prove the initial claim that the students can complete the tangram in less than 2 minutes, even in the face of the arithmetical average of the samples $\bar{x}$ being less than 120.
This might come as a surprise, however, having a high confidence demand of 95% also plays a role in this result.

Appendices

The original data, the scripts for the filtering of data, and the excel sheet are provided under the following GitHub repository: https://github.com/A01831983/M2-Act1

However, this result likely can not be generalised to all the students at Tec, since with such a sample size, the central limit theorem would gain much more accurate approximations.
Also, such a more diverse pool of subjects might unveil biases in our subjects, which remain hidden for now (e.g., students coming from similar courses of study).